

MemoryOverwrite

A cybersecurity game



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01 Game Concept

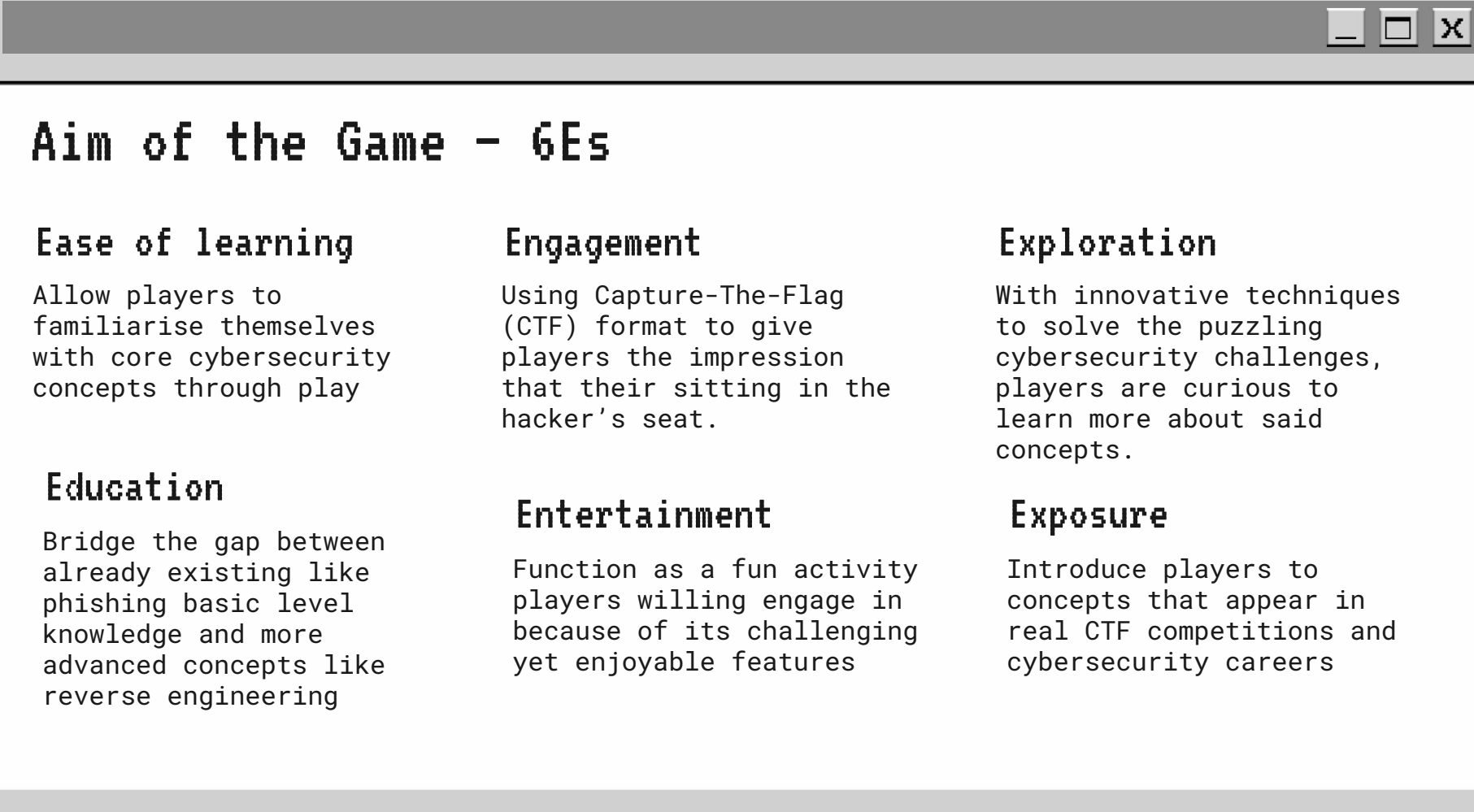


Trapped inside a virtual world with your memory wiped, Memory0verwrite turns you into the hacker, weaponizing real cyberattacks to outsmart a rogue virus and break free.

02 Aim of the Game



=) 6Es



Aim of the Game - 6Es

Ease of learning

Allow players to familiarise themselves with core cybersecurity concepts through play

Education

Bridge the gap between already existing like phishing basic level knowledge and more advanced concepts like reverse engineering

Engagement

Using Capture-The-Flag (CTF) format to give players the impression that their sitting in the hacker's seat.

Entertainment

Function as a fun activity players willing engage in because of its challenging yet enjoyable features

Exploration

With innovative techniques to solve the puzzling cybersecurity challenges, players are curious to learn more about said concepts.

Exposure

Introduce players to concepts that appear in real CTF competitions and cybersecurity careers

03 Target Audience



Designed for tech-curious teens (ages 13-17) and total beginners, this game is the ultimate, zero-coding-required launchpad for future ethical hackers.

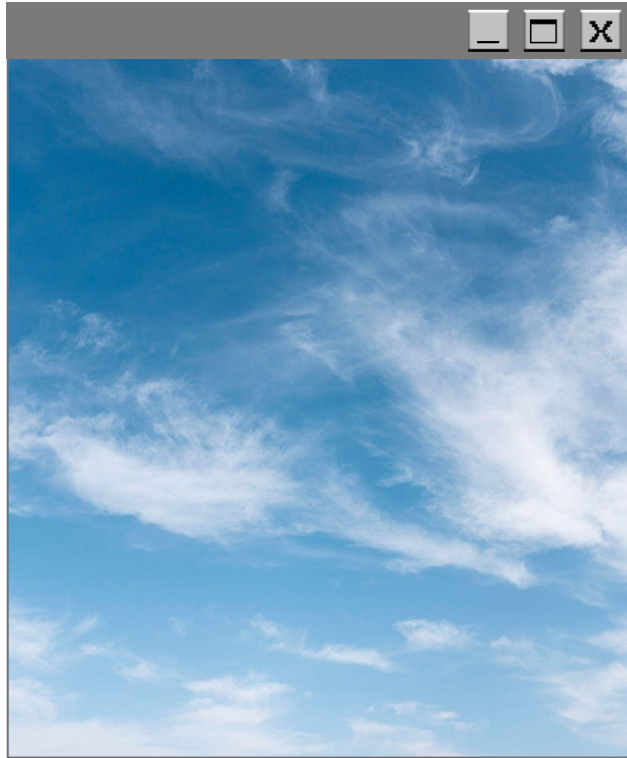
04 Game Features



Game Features

Progression

- 8 themed rooms arranged in 4 difficulty tiers - players must clear each room to unlock the next
- Locked rooms visible on the world map from the start, showing titles and preview descriptions to build curiosity
- Revisitable rooms: players can return to any cleared room to re-attempt for a better score or explore bonus content



But you must be
curious about what
EXACTLY do we have in
stock.

So let us introduce our...



Room Roster

Tier 1 - Entry Point (*Basic intuitive knowledge*)

Room 1 - Phishing Lab

Concept: Social engineering via email

Mechanic:

- An inbox displays 5-8 simulated emails, each fully rendered with sender, subject, and body
- Player clicks emails to inspect it and marks it as "Safe" or "Phishing"
- Submit all answers to clear the room

Tier 1 - Entry Point (*Basic intuitive knowledge*)

Room 1 - Phishing Lab

Concept: Social engineering via email

What makes an email suspicious:

- Misspelled sender domain (e.g. [paypal.com](#))
- Urgency language ("Act NOW or your account closes")
- Mismatched hover URL vs. displayed link text
- Generic greeting ("Dear User")
- Unexpected attachment



Tier 1 - Entry Point (*Basic intuitive knowledge*)

Room 1 - Phishing Lab

Concept: Social engineering via email

Exploration bonus: Hover tooltips in "analyst mode" explain each red flag.
A collectible "hacker's notes" entry explains the psychology behind phishing - authority, urgency, fear.

What the player learns by doing: How to read an email critically - sender inspection, URL analysis, urgency tactics, social manipulation cues.

Tier 1 - Entry Point (*Basic intuitive knowledge*)

Room 1 - Phishing Lab

Concept: Social engineering via email

Feedback system: Wrong answers highlight the specific clue missed with a one-line explanation - not just "wrong" but *why* it was wrong.

Tier 1 - Entry Point (*Basic intuitive knowledge*)

Room 2 - OSINT Investigator (INCLUDED IN DEMO)

Concept: Open-source intelligence gathering

Mechanic:

- A simulated social media + forum environment
- Player is given a target username and must piece together a profile using only publicly visible data scattered across fake profiles
- Final objective: build a complete dossier (name, employer, location, daily routine)

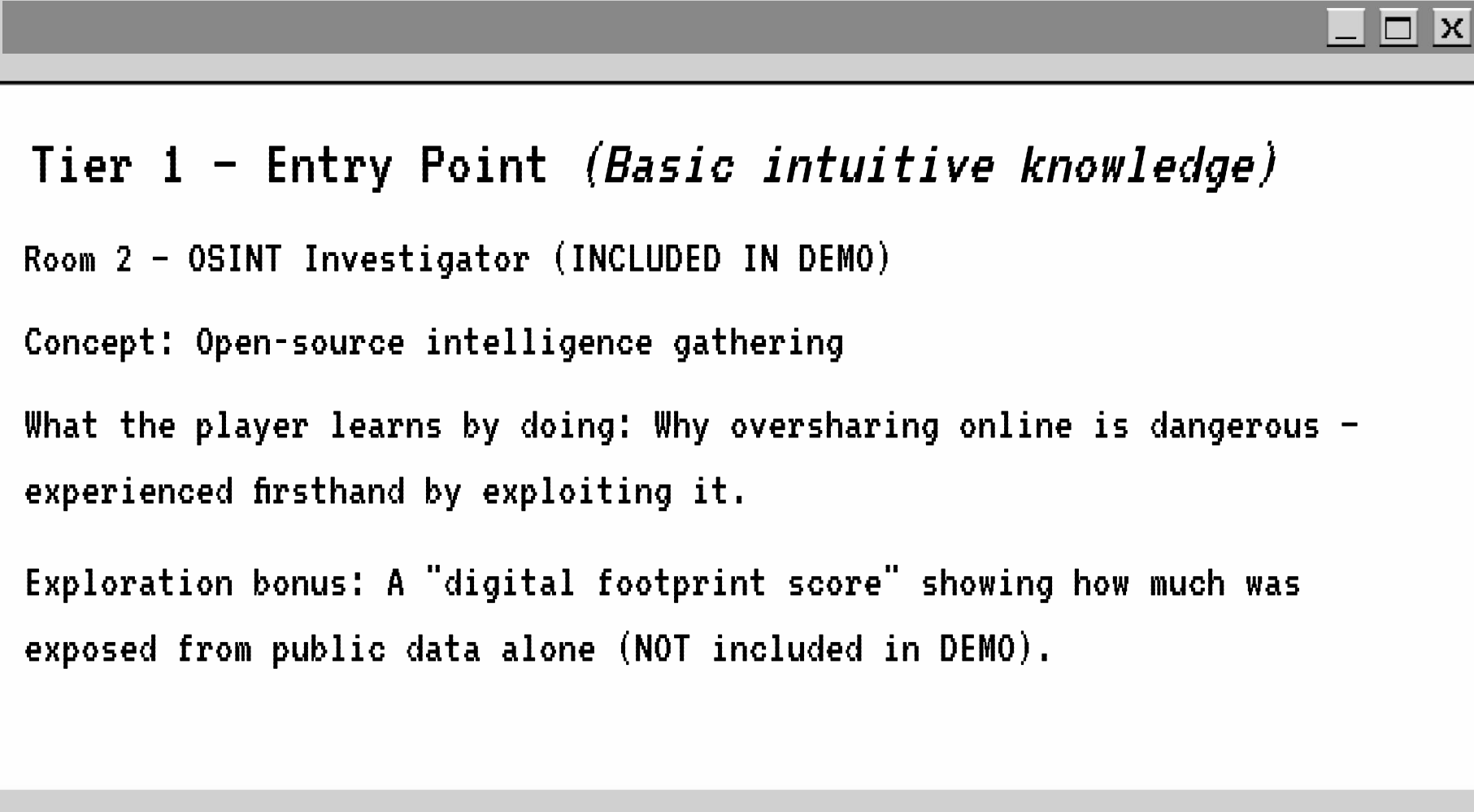
Tier 1 - Entry Point (*Basic intuitive knowledge*)

Room 2 - OSINT Investigator (INCLUDED IN DEMO)

Concept: Open-source intelligence gathering

Puzzle clues:

- Photo metadata reveals a location
- Forum post timestamps reveal a timezone
- Username reuse across platforms connects accounts
- A pet's name appears - and becomes a usable password in Room 3



Tier 1 - Entry Point (*Basic intuitive knowledge*)

Room 2 - OSINT Investigator (INCLUDED IN DEMO)

Concept: Open-source intelligence gathering

What the player learns by doing: Why oversharing online is dangerous - experienced firsthand by exploiting it.

Exploration bonus: A "digital footprint score" showing how much was exposed from public data alone (NOT included in DEMO).

Tier 2 - Getting Technical (*Light pattern recognition*)

Room 3 - Password Vault (INCLUDED IN DEMO)

Concept: Password cracking, hashing, salting

1. Dictionary attack - crack a weak password by selecting words from a wordlist; watch attempts tick through
2. Hash cracking - given an MD5-style hash and a mini rainbow table, match the hash to its plaintext password
3. Salting demo - same password produces a different hash; player figures out why
4. Fortify it - build a new strong password; a live strength meter responds to input; the same dictionary attack runs again and fails



Tier 2 - Getting Technical (*Light pattern recognition*)

Room 3 - Password Vault (INCLUDED IN DEMO)

Concept: Password cracking, hashing, salting

Exploration bonus: A live "time to crack" display - updates from "0.003 seconds" to "centuries" as the password improves. Famous bad passwords (123456, qwerty, password) crack instantly with a "used in X million breaches" flavour note.

What the player learns by doing: Why weak passwords fail (by cracking one), what hashing means, why salting matters, how attackers approach credential attacks.

Tier 2 - Getting Technical (*Light pattern recognition*)

Room 4 - Cipher Chamber

Concept: Encryption fundamentals - Caesar, Vigenère, XOR

Mechanic (three stages):

1. Caesar cipher - encrypted message on screen; player adjusts a shift slider (1-25); text decodes live
2. Key-based cipher - a keyword must be found by exploring the room environment; fragments hidden as clues
3. XOR puzzle - two bit-strings displayed side by side; player toggles bits to apply the key; encrypted message decodes in real time

Tier 2 - Getting Technical (*Light pattern recognition*)

Room 4 - Cipher Chamber

Concept: Encryption fundamentals - Caesar, Vigenère, XOR

Why XOR matters: XOR is the actual foundation of modern encryption and appears constantly in CTF challenges. Showing the bitwise operation visually (not algebraically) makes it accessible.

Exploration bonus: A frequency analysis tool - shows letter frequency in the ciphertext. Advanced players can crack the Caesar cipher analytically rather than brute-forcing the slider. Labelled explicitly as "what CTF players actually use."



Tier 2 - Getting Technical (*Light pattern recognition*)

Room 4 - Cipher Chamber

Concept: Encryption fundamentals - Caesar, Vigenère, XOR

What the player learns by doing: How encryption obscures information, how keys control access, why key length and complexity resist brute force - all through interaction, not instruction.

Tier 3 - Hacker Brain (*System-thinking, Code concepts*)

Room 5 - Network Recon

Concept: Port scanning, service enumeration, attack surface mapping

Mechanic:

- Player operates a simulated nmap-style scanner
- Sees a visual network of nodes
- Runs "scans" by selecting scan type and target - results populate a table of open ports and services
- Must identify which service is vulnerable from the scan results and plot the attack path

Tier 3 - Hacker Brain (*System-thinking, Code concepts*)

Room 5 - Network Recon

Concept: Port scanning, service enumeration, attack surface mapping

Phases:

1. Ping sweep - find live hosts on the subnet
2. Port scan - discover open ports per host
3. Service detection - identify what's running on each port
4. Vulnerability identification - match service version to a known CVE (simplified)
5. Attack path planning - route through the network to the target

Tier 3 - Hacker Brain (*System-thinking, Code concepts*)

Room 5 - Network Recon

Concept: Port scanning, service enumeration, attack surface mapping

1. Engagement hook: Scan results animate in like a real terminal - text streaming line by line. Hovering a node shows its IP, open ports, and role (router, server, endpoint).
2. Exploration bonus: A "traceroute" view showing the full path taken after completing the room - mirrors the real `traceroute` command.
3. What the player learns by doing: How data moves through networks, what routers and firewalls do, how ports control traffic, enumeration as a discipline.

Tier 3 - Hacker Brain (*System-thinking, Code concepts*)

Room 6 - Script foundry

Concept: Python basics applied to hacking automation

Mechanic:

- An in-game code editor (simulated terminal - no real Python runtime required)
- Player is given partially written scripts with gaps; fill in the blanks to make the script work
- When the script runs correctly, it produces a visible in-game effect

Tier 3 - Hacker Brain (*System-thinking, Code concepts*)

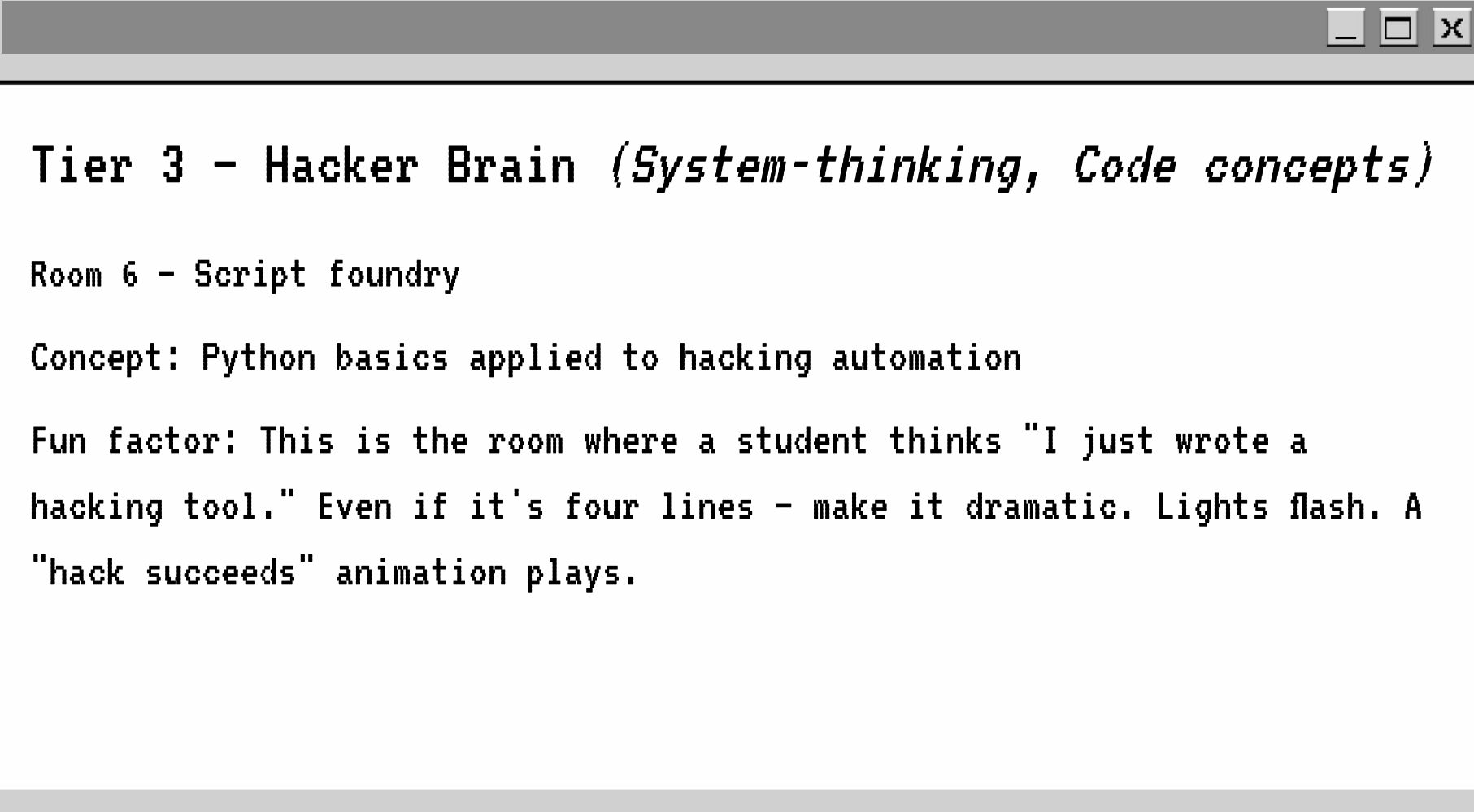
Room 6 - Script foundry

Concept: Python basics applied to hacking automation

Why fill-in-the-blank: Writing code from scratch is too high a bar for beginners.

Completing a working script that produces a visible result gives the same "I made this work" moment with a much lower entry cost. Scaffolded programming is proven pedagogy.

What the player learns by doing: Basic Python syntax and logic, how automation amplifies attack capability, the connection between code and real-world security tools.



Tier 3 - Hacker Brain (*System-thinking, Code concepts*)

Room 6 - Script foundry

Concept: Python basics applied to hacking automation

Fun factor: This is the room where a student thinks "I just wrote a hacking tool." Even if it's four lines - make it dramatic. Lights flash. A "hack succeeds" animation plays.

Tier 4 - Deep Dive (*Advanced - for the curious*)

Room 7 - Reverse Engineer

Concept: Reading assembly/pseudocode to understand program logic

Mechanic:

- A fake disassembler shows simplified pseudocode/assembly of a "locked program"
- Player reads the logic to determine what input will print "ACCESS GRANTED"
- No prior assembly knowledge needed - guided tooltips explain registers, comparisons, and jumps contextually

Tier 4 - Deep Dive (*Advanced - for the curious*)

Room 7 - Reverse Engineer

Concept: Reading assembly/pseudocode to understand program logic

What the player learns by doing: That the answer is already inside the program - you just have to read it backwards. The core insight of reverse engineering, achievable with zero assembly background.

Tier 4 - Deep Dive (*Advanced - for the curious*)

Room 7 - Binary breach (BBEG Room)

Concept: Buffer Overflow, Memory corruption,

What the player learns by doing: That the answer is already inside the program - you just have to read it backwards. The core insight of reverse engineering, achievable with zero assembly background.

05 Why it's fun
AND useful



Why is it fun & useful?

Fun

- Each “room” has a distinct mechanic which requires different perspectives each to solve.
- The game plays out from a Hacker’s perspective, adding more depth to the game rather than a basic cybersecurity game

Useful

- Students learn through repeated trial and error rather than merely being told the correct answer.
- It serves as a good start for students who want to understand cybersecurity but are facing difficulties

06 Design Choices



You can enter a subtitle here if you need it

Design Choices

Hacker's Perspective

Players' UX from the hacker's perspective teaches to attack first so they know how to adequately defend themselves

Tier-Based

Increases in difficulty from
"room" to "room"

OK

Cancel

Apply

Design Choices

Locked “rooms”

All “rooms” will be visible from the room select menu but will appear to be locked for Player suspense

Narrative payoff

The final room of the game resolves the story defeating the BBEG

OK

Cancel

Apply

Design Choices

CTF format

The game UI gives of a similar feel to CTF competitions giving players a more Hacker like impression

Immersive Play

Players will feel as if they are the ones interacting with the computer in game through first person view

OK

Cancel

Apply

07

Future Developments





Future developments



More "Rooms"

Other rooms with concepts like Web exploitation, Forensics and Malware analysis can be added to the game

New BBEG

With more "rooms", new BBEGs can be introduced to the game

Multiplayer

Incorporate co-op mode or squad mode to work as a team and solve the challenges

Ranked Matches

Compete against other players, with ranked matches, climb the leaderboard and earn rewards for your rank

Future developments



Teacher

A classroom mode where educators can utilise the game to teach students as well as a platform where players can ask how to beat certain rooms



CTF integration

The game can have a authenticated platform through which it can export a "CTF Ready" certificate listing allowing them to participate in beginner CTF platforms



Scripting

Players will be allowed to write full on scripts to receive better rewards rather than continuing with fill-in-the-blank answers already existing in the current game



Leaderboard

Implementing achievements and rewards or penalties for timed runs, hints used and for exploration to add a more competitive feel as well as encourage replays.

Write your Name



SYSTEM RESTORED

VIRUS DEFEATED // ALL MEMORY FRAGMENTS RECOVERED

ROOMS CLEARED

2 / 2

MEMORY FRAGMENTS

2 / 2

TOTAL SCORE

800 pts

COMPLETION

100%

[RECOVERED MEMORY FRAGMENTS]

Write your Name



Thank you

Do you have any questions?



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OK

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Claude code	Vibe coding	-
Claude code	Debugging	-